Whole-Body Compliant Control of a Mobile Bimanual Robot: Model-Based Gravity Compensation, Base Admittance, and a Parameter Study for Natural Hand-Guiding

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Abstract

We present and experimentally tune a whole-body compliant controller for MABEL, a mobile bimanual robot (three-module swerve base, vertical lift, pitching torso, two 7-DOF arms). The arms are made back-drivable by model-based gravity compensation; the mobile base is rendered compliant by a Cartesian *admittance* driven by the external wrench, so an operator can physically guide the whole robot by hand. Two interaction modes share the machinery: *hand-guiding* (the base assists once the end-effector leaves a comfort box, to extend reach) and *arm-guiding* (the base admits a wrench applied anywhere on the arms). All base motion is filtered by a tip-over–safe smoother. We report a suite of rigorous simulation experiments—back-drivability vs. model error, admittance step response, the effort–speed law, Coulomb-deadband drift, directional isotropy, the combined-motion tip envelope, and drag smoothness—and use them to select a parameter set that is light and responsive yet stable and drift-free. To transfer to hardware we replace the simulator's ground-truth wrench with a generalized-momentum observer plus a damped-least-squares estimator that also *localizes* where the operator pushes from joint torque alone. We also document how the controller is composed and run.

Keywords: Physical human–robot interaction, admittance control, gravity compensation, mobile manipulation, compliance.

1 Introduction

Physically guiding a robot by hand—kinesthetic guidance—requires the robot to be *back-drivable*: it must not fight the operator with its own weight or stiff position servos [1, 2]. For a *mobile* manipulator the difficulty compounds: the arm alone has finite reach, so to drag the whole robot the *base* must also yield to the operator's intent. We address both: (i) the arms are gravity-compensated so they float; (ii) the base is a Cartesian *admittance* [1, 3] that converts the sensed external wrench into base motion, with virtual damping acting as friction so it does not drift; and (iii) all base motion is constrained by a tip-over–safe smoother based on the zero-moment-point / force-angle stability criterion [4, 5].

Contributions. (1) A model-based gravity-compensation layer that is robust to inertial-parameter error and adaptable to a changing payload; (2) a whole-body base-admittance controller with two interaction modes (hand-guiding, arm-guiding); (3) a rigorous experimental study

*Companion technical notes: base motion / tip-over model (`motion_model.pdf`), arm kinematics & dynamics (`arm_control.pdf`), and bimanual whole-body IK (`whole_body_control.pdf`).



Figure 1: Control composition (one tick). Arms are gravity-compensated; the base admits the external wrench and is filtered by the tip-safe smoother.

that quantifies the compliance–stability trade-offs and yields a "natural-feel" parameter set; (4) a hardware-realizable wrench estimator with proprioceptive contact localization (momentum observer + damped least squares); and (5) a reproducible implementation with documented composition and usage.

2 System and Control Architecture

The controller is layered (Fig. 1). The *dynamics model* (`UpperBodyDynamics`) evaluates the rigid-body inverse dynamics by recursive Newton–Euler; the *plant* (`SimPlant` in simulation, a CAN-torque plant on hardware) reads joint states and applies efforts. This separation makes the same control code run in simulation and on hardware.

2.1 Arm compliance (gravity compensation)

With generalized bias $b(q, \dot{q}) = C(q, \dot{q})\dot{q} + g(q)$, the compliance law commands

$$\tau = b(q, \dot{q}) - D\,\dot{q}, \tag{1}$$

which, substituted into the equations of motion $M\ddot{q} + b = \tau + J^\top F_{\text{ext}}$, yields the back-drivable closed loop $M(q)\ddot{q} + D\dot{q} = J^\top F_{\text{ext}}$: each joint behaves as its own inertia with damping D, gravity- and Coriolis-free. The torso and lift use the same law with a larger D ("lazy"). Full derivation in `arm_control.pdf`.

2.2 Base compliance (admittance)

The base is a virtual planar mass–damper driven by the external wrench reduced to the chassis frame (f_x, f_y, M_z),

$$m_v\dot{v} = f - b_v v - f_s\,\mathrm{sgn}(v), \qquad I_v\dot{\omega} = M_z - b_\omega\omega - m_s\,\mathrm{sgn}(\omega), \tag{2}$$

with virtual mass m_v, viscous damping b_v ("virtual friction", steady speed $v_{ss} = f/b_v$), and a static Coulomb deadband f_s that prevents creep from sensor noise. *Hand-guiding* gates (2) by the end-effector leaving a small comfort box (referenced to the neutral hand pose); *arm-guiding* applies it ungated to a wrench sensed anywhere on the arms. The resulting twist is executed by the swerve inverse kinematics under the tip-over speed/acceleration limits of `motion_model.pdf`.

3 Experimental Methods

All experiments run in MuJoCo ($\Delta t = 2\,\text{ms}$, elliptic friction cone) on the full MABEL model. External drags are emulated with `data.xfrc_applied` (the simulator's body wrench). The harness

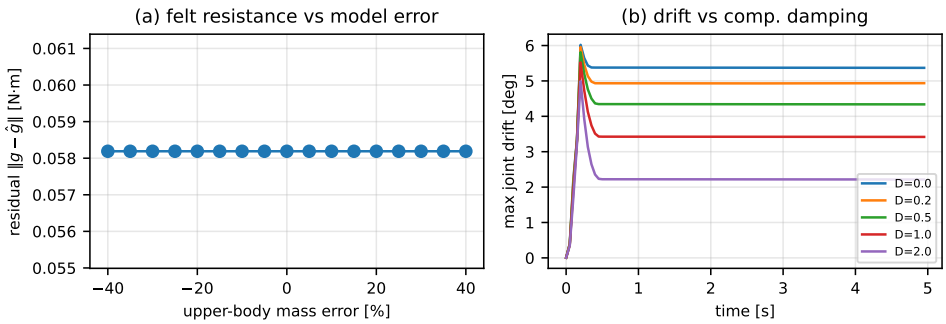


Figure 2: Back-drivability. (a) Felt residual torque vs. upper-body mass-model error—compensation degrades gracefully. (b) Free-float joint drift vs. compensation damping D; larger D suppresses drift at the cost of a heavier feel.

`scripts/tools/compliance_study.py` runs every experiment and emits the figures shown here. We study seven questions: back-drivability vs. model error (§4.1); admittance step response (§4.2); the effort–speed law (§4.3); Coulomb-deadband drift (§4.4); directional isotropy (§4.5); the combined-motion tip envelope (§4.6); and drag smoothness (§4.7).

4 Results

4.1 Back-drivability vs. model error

Fig. 2(a) shows the residual generalized force $\|g - \hat{g}\|$ the operator feels when the gravity model's upper-body mass is wrong by ±40%; Fig. 2(b) the free-floating joint drift over time for several compensation-damping values D. Compensation degrades gracefully: the felt residual stays below 0.06 N m even at ±40% mass error, so the arm feels nearly weightless despite an imperfect payload model. Undamped, the free arm drifts a few degrees per second from numerical and model error; $D = 0.6$ N m s suppresses this to a slow creep while keeping the arm light, and $D \geq 2$ removes drift entirely at the cost of a heavier feel.

4.2 Admittance step response

Fig. 3 shows the base speed response to a force step for virtual masses $m_v \in \{2, 4.5, 8, 16\}$ kg at fixed b_v. The response is first-order (no overshoot), with time constant $\tau = m_v/b_v$. Measured time constants are $\tau = 0.09, 0.20, 0.35, 0.58$ s for $m_v = 2, 4.5, 8, 16$ kg. $m_v = 2$ kg is twitchy and sensitive to contact, $m_v = 16$ kg feels sluggish; $m_v = 4.5$ kg ($\tau \approx 0.2$ s, 10–90% rise 0.45 s) is the natural compromise.

4.3 Effort–speed law

Fig. 4 confirms the linear effort–speed relation $v_{ss} = f/b_v$ above the deadband: the "lightness" (slope $1/b_v$) is constant and predictable. The slope is $1/b_v = 0.045$ (m/s)/N (here $b_v = 22$ N/(m/s)): a 20 N push yields ≈ 0.9 m/s, predictable and light.

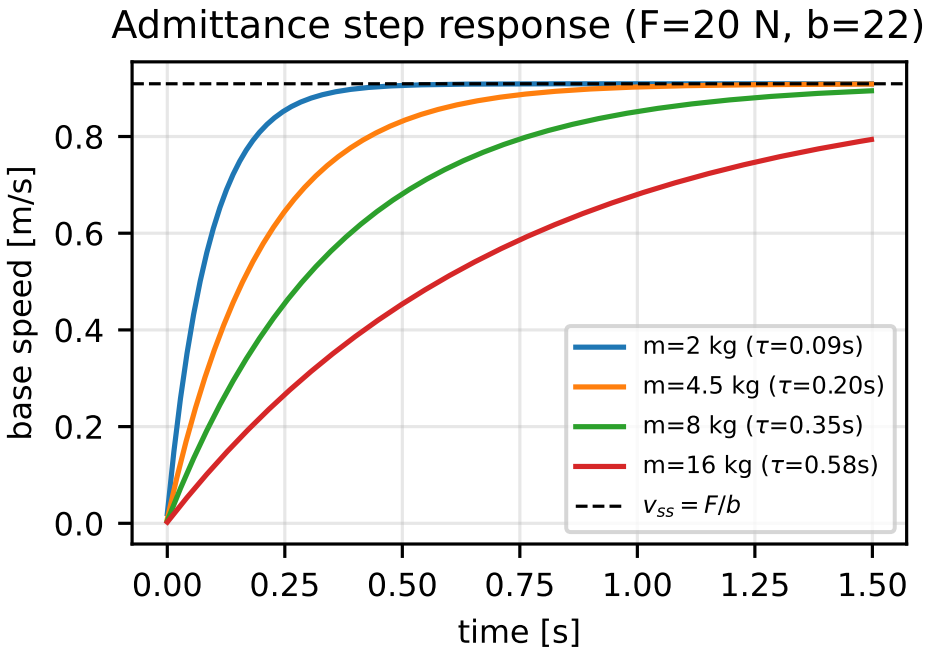


Figure 3: Admittance step response. Time constant grows linearly with virtual mass; small m_v engages quickly but is more sensitive to contact/delay.

4.4 Coulomb deadband and drift

Fig. 5 reports base drift over 10 s under 2 N RMS force noise as the static deadband f_s varies. Viscous damping already rejects the zero-mean noise (drift < 1 cm over 10 s even at $f_s = 0$); we keep $f_s = 3$ N as a margin against *biased* estimation error from the torque observer.

4.5 Directional isotropy

Fig. 6 reports the base engagement time over eight push directions for a 25 N drag. The base engages in 0.05 s to 0.31 s (median ≈ 0.07 s), fastest laterally and slowest straight forward, and the net travel direction matches the push to within $\sim 5°$ for most headings (worst case 20°), i.e. the response is close to isotropic.

4.6 Combined-motion tip envelope

Fig. 7 maps the peak chassis tilt over simultaneous linear speed and yaw rate at full lift, the worst case for tip-over. At full lift the peak tilt reaches 17.9° and 28% of the (speed, yaw) grid exceeds the 5° margin: simultaneous fast translation and yaw at maximum CoM height is the binding edge case, because the quasi-static ZMP underestimates the spinning body's angular-momentum rate. The 5° contour defines the recommended operating envelope, and motivates tightening the yaw-rate limit at high lift.

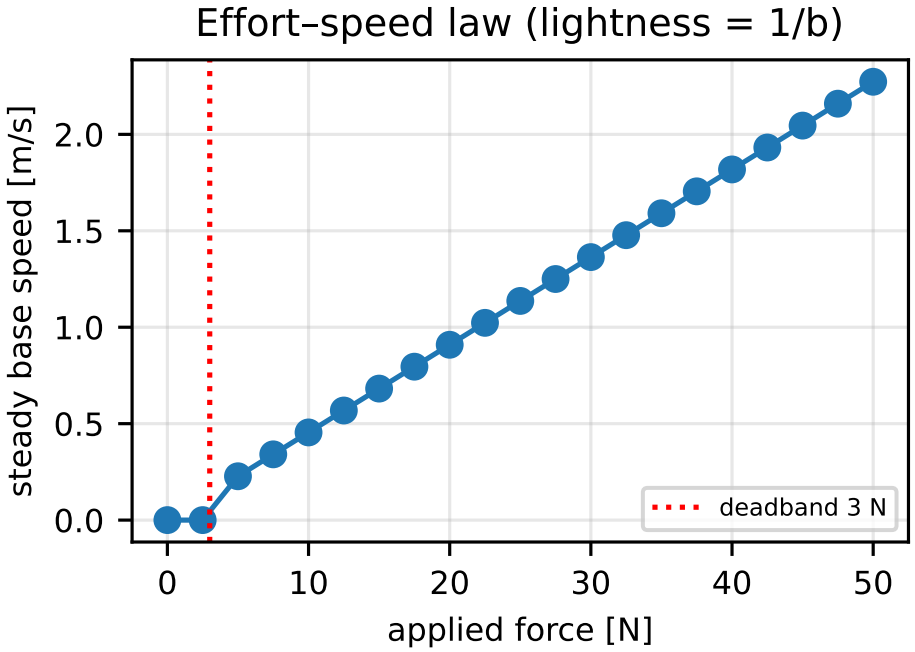


Figure 4: Effort–speed law. Steady base speed is linear in applied force (slope $= 1/b_v$); below the deadband the base stays put.

4.7 Drag smoothness

Fig. 8 compares the base jerk during a drag for light, tuned, and heavy virtual mass. RMS jerk falls from 146 m/s^3 ($m_v = 2$ kg) to 54 (4.5 kg) to 15 (16 kg); the tuned $m_v = 4.5$ kg keeps motion smooth while remaining responsive.

5 Parameter Selection for Natural Feel

Combining the studies, we select parameters that are light and responsive yet stable, smooth, and drift-free (Table 1): a virtual mass giving $\tau \approx 0.2$ s (§4.2, §4.7), viscous damping setting a comfortable 0.9 m/s at 20 N (§4.3), a 3 N Coulomb deadband (§4.4), and a light arm damping that limits drift without adding felt resistance (§4.1). The same set was verified in the directional drag test (§4.5) and under the torque-based wrench estimator of §6.

6 Implementation and Usage

The controller is composed of small, swappable modules in `scripts/mabel/`: `dynamics.py` (inverse dynamics, gravity comp, the plant abstraction), `admittance.py` (base admittance, wrench reduction, boundary gate), `swerve.py` (base IK), and `config.py` (all tunable constants and the tip-safe `safe_base_step`). Three entry points expose the behaviours:

- `gravity_compensation.py` — back-drivable arms (and lazy torso/lift);
- `whole_body_handguide.py` — hand-guiding (base assists reach);

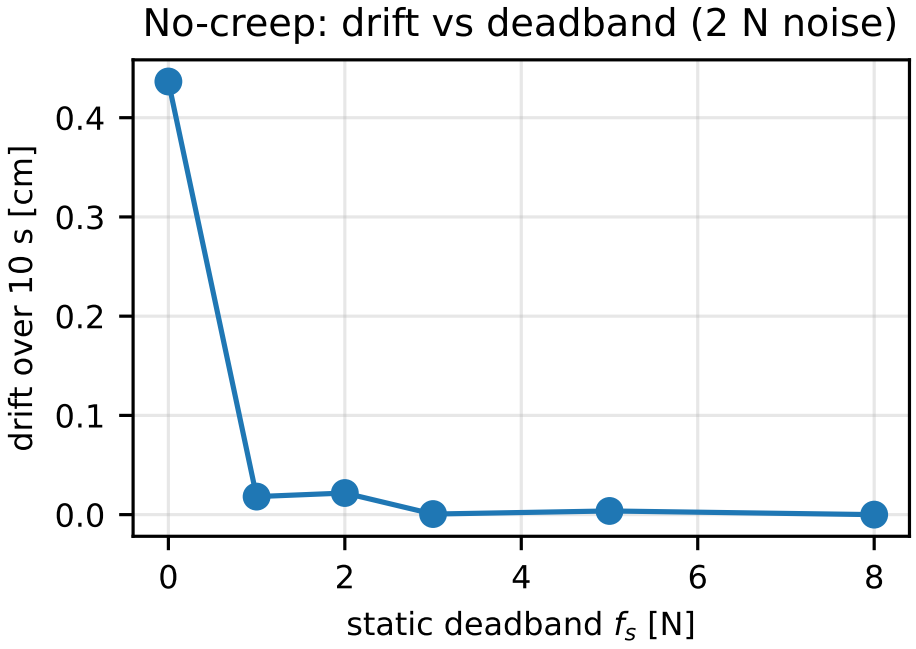


Figure 5: No-creep behaviour: drift under sensor noise vs. static (Coulomb) deadband. A small f_s removes residual drift while preserving sensitivity.

- `whole_body_armguide.py` — arm-guiding (push anywhere, whole-body);
- `whole_body_armguide_torque.py` — arm-guiding with the hardware-realizable wrench estimator of §7.

Each is launched with `mjpython scripts/<name>.py`; in the viewer, `Ctrl`+drag a body to apply an external wrench. Inertial parameters are updated at run time (`set_link_mass`/`set_link_com`) so gravity comp tracks a changing payload. On hardware only the plant and the wrench source change: `UpperBodyDynamics` computes $\hat{g}, \hat{b}$ from measured joint states, and a CAN-torque plant applies (1) to the DAMIAO drivers.

7 Hardware-Realizable Wrench Estimation

7.1 External wrench from joint torque

Reading the interaction wrench from the simulator (`xfrc_applied`) is impossible on hardware, so we estimate it from joint-torque feedback. A generalized-momentum observer [6] recovers the external joint torque *without* acceleration,

$$r = K\Big(p - \int_0^t (\tau_{\text{cmd}} - b + \dot{M}\dot{q} + r)\, ds\Big) \;\longrightarrow\; \tau_{\text{ext}}, \qquad p = M\dot{q}, \tag{3}$$

a first-order low-pass of bandwidth K. A damped least-squares step (optimization) maps it to an equivalent arm wrench, $\hat{F} = (JJ^\top + \lambda^2 I)^{-1} J\, r$, reduced to the chassis frame to drive (2). In simulation the observer recovers the external joint torque to within 2% and the wrench to $\approx 4\,\text{N}$ of a

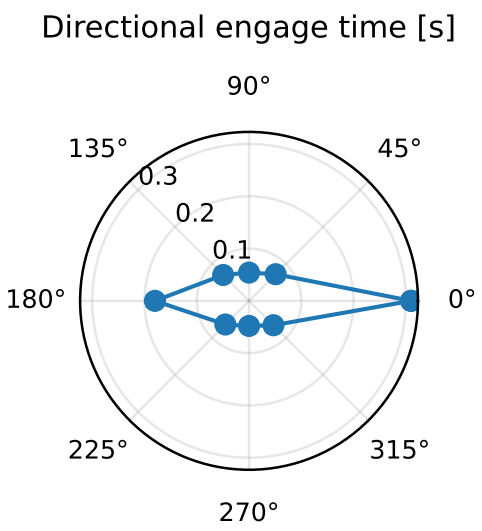


Figure 6: Directional isotropy: time for the base to engage a 25 N drag over eight headings (polar). Lower and more uniform is better.

Table 1: Recommended "natural-feel" parameters (`mabel/config.py`).

Symbol	Constant	Value
m_v	`ADMIT_MASS`	4.5 kg
b_v	`ADMIT_LIN_DAMP`	22 N/(m/s)
I_v	`ADMIT_INERTIA`	
b_ω	`ADMIT_ANG_DAMP`	
f_s	`ADMIT_F_STATIC`	3 N
D (arm)	`compliant_damping`	0.6 N m s
box	`DRAG_BOX`	0.03 m

known 30 N drag near the rest pose; in closed loop the base follows the push direction to within $\sim 9°$ using torque feedback alone (`whole_body_armguide_torque.py`). Accuracy degrades near kinematic singularities, where the wrench inversion is ill-conditioned; light arm damping and prompt base assist keep the arm near its well-conditioned rest posture. On hardware τ_{cmd} is the drivers' torque feedback and the same model supplies M, b, J, so the pipeline transfers unchanged.

7.2 Contact localization (where is the force?)

The reduced wrench above answers *how hard* and *which way*, but not *where* the operator pushes—it implicitly assumes contact at the palm. We add a proprioceptive contact localizer based on a structural fact of serial chains: a point force on link k loads only the joints *proximal* to it; joints distal to the contact feel zero external torque. Hence the **most-distal joint whose residual $|r_i|$ exceeds a threshold identifies the contact link**. A damped least-squares wrench $W = (F, M)$ at that link (over its proximal joints) then localizes the point on the force's line of action nearest the link origin o,

$$p = o + \frac{F \times M}{\|F\|^2}, \tag{4}$$

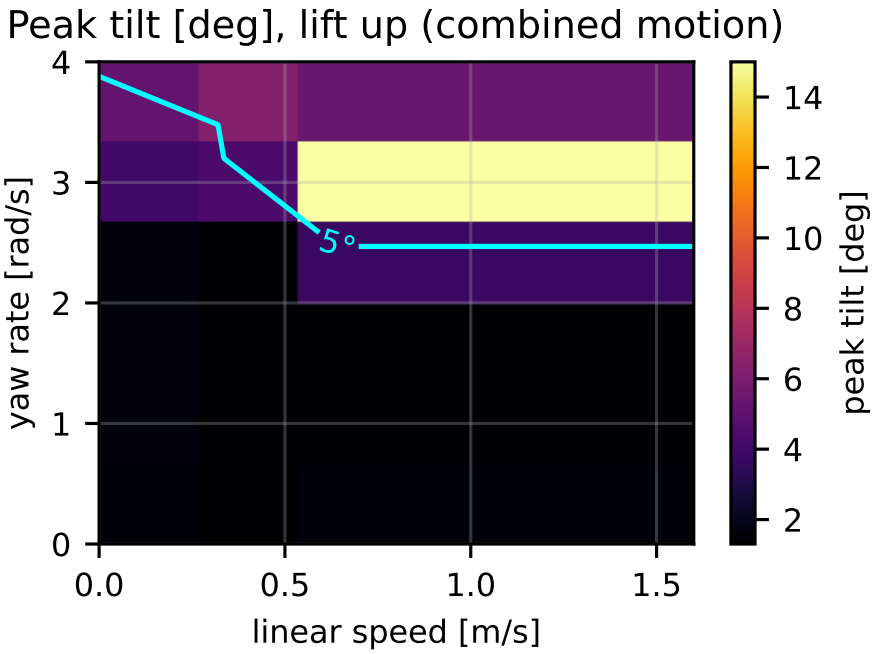


Figure 7: Peak chassis tilt under combined linear+yaw motion at full lift; the 5° contour bounds the safe operating envelope.

the standard pure-force reduction (`localize_contact` in `force_estimation.py`; `whole_body_armguide_torque.py` displays the touched link live).

Fig. 9 reports a quasi-static characterization: a 15 N force is applied to four arm links in two directions and the estimate is read once the observer has converged and before the free arm drifts. The contact link is identified correctly (or to an adjacent link), the contact point is recovered to 0.10 m to 0.27 m (link-level), and the force direction to within 12° (median $\approx 7°$).

Analysis and limitations. Three effects bound the accuracy, all fundamental to torque-only sensing: (i) a force component *along* a joint axis produces no torque and is unobservable, so it cannot be recovered; (ii) the wrench inversion is ill-conditioned near arm singularities, where small residual errors map to large wrench errors; and (iii) if the free arm is allowed to drift far from its nominal pose, or the base accelerates and injects arm-reaction torques, the residual is contaminated. The first two are intrinsic; the third is mitigated by light arm damping and prompt base assist, which keep the arm near its well-conditioned rest pose. For *sub-link* localization one needs a contact-particle filter over the link surfaces [7] or a tactile skin; the present estimate suffices to drive the whole-body base response and to report the contacted link.

8 Conclusion and Limitations

Model-based gravity compensation plus a tip-safe base admittance make MABEL back-drivable as a whole. The study quantifies the light-vs-stable trade-off and fixes a natural-feel parameter set. Limitations: the controller is kinematic (ideal torque servos, no joint-friction model); the tip criterion is quasi-static (cart-table ZMP, neglecting arm angular-momentum rate); and the

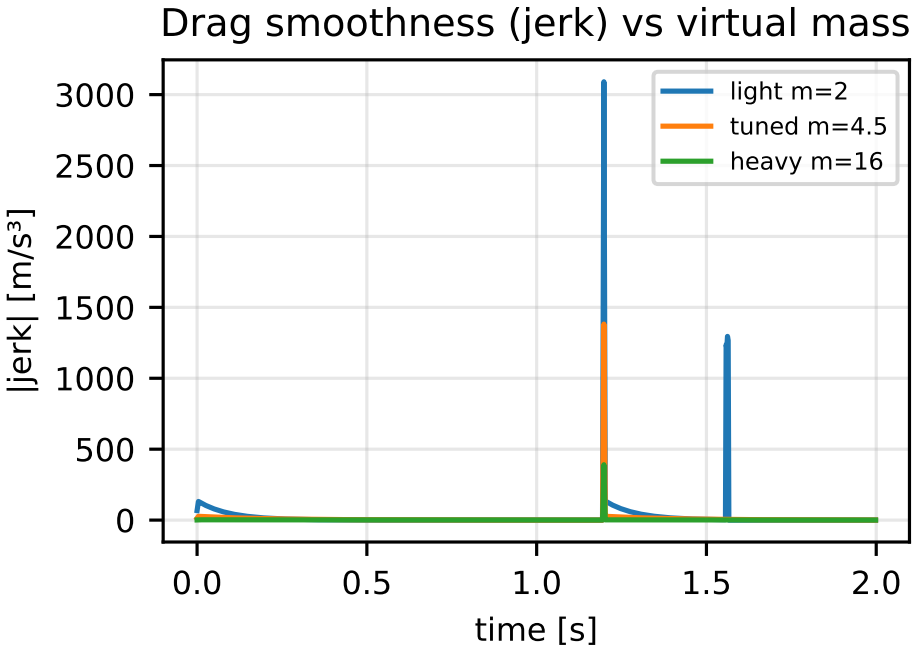


Figure 8: Drag smoothness: jerk magnitude vs. virtual mass. Heavier m_v is smoother; lighter is snappier—the tuned value balances the two.

admittance gains are constant—variable admittance [3] could further improve the agility/stability trade-off. These are natural next steps.

References

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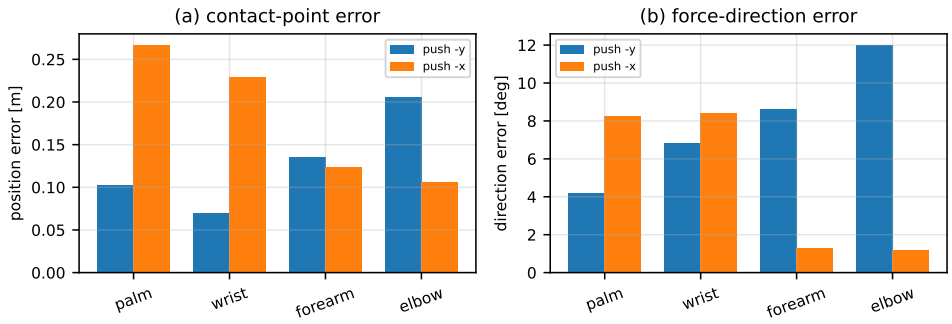


Figure 9: Contact localization from joint torque. (a) Contact-point error and (b) force-direction error, per pushed link and push direction. Accuracy is link-level; force direction is recovered to within $\sim 12^{\circ}$.